and Neuro-Symbolic approaches to combine LLMs with cognitive architectures [174, 175]. These projects align with the CMC, integrating cognitive architectures like ACT-R, Soar, and Sigma to provide a unified framework for human cognition [178]. Enhancements in AI robustness and interoperability were achieved by integrating cognitive architectures with foundation models for cognitively guided few-shot learning [167]. Finally, combining generative networks with the CMC using a Neuro-Symbolic approach merged symbolic reasoning with neural networks to replicate human cognitive processes for powerful, explainable AI systems has been theorised but not yet realised[176]. Open research questions remain around how Neuro-Symbolic AI can integrate symbolic reasoning with meta-reinforcement learning for complex decision-making, fuse cognitive architectures with LLMs to develop meta-cognitive agents, leverage LLMs to enhance instance-based learning through meta-cognitive signals, create adaptive meta-cognitive frameworks for real-time conflict resolution, combine modular and agency approaches to build meta-cognitive AI systems aligned with the Common Model of Cognition, improve few-shot learning with cognitive architectures for meta-cognitive awareness, and develop Neuro-Symbolic generative networks that replicate human-like meta-cognitive processes.

5. Meta-Cognition in Neuro-Symbolic AI

Whilst the initial representation of Neuro-Symbolic AI as system 1 and system 2 level thinking is a useful tool to goal-orientate the field towards a common direction for the integration of neural and symbolic processes, the current adaptation of the human-level cognitive processing ability is too simplistic and does not yet capture the full systems-level breakdown of where the community should be investing effort to push the field forward. As Kahneman himself states, “the two systems do not really exist in the brain or anywhere else. ’System 1 does X’ is a shortcut for ’X occurs automatically.’ And ’System 2 is mobilized to do Y’ is a shortcut for ’arousal increases, pupils dilate, attention is focused, and activity Y is performed.’”. Human-level cognition manifests from a deeply complex and intricately layered yet densely connected box of systems of systems that work in unification and act with system-1 system-2 characteristics. The goal of the Neuro-Symbolic research domain is to “create a superior hybrid AI model possessing reasoning capabilities” and hence, to push the field further towards this goal we must seek to design and build systems that act with the same propensity as Kahnemans system-1 system-2 character setup through the implementation of more integrated systems of systems controlled through Meta-Cognition possessing the ability to act lazily when necessary and focused when required.

6. Conclusion

The field of Neuro-Symbolic AI has experienced a notable surge in research activity from 2020 onwards, reflecting the growing recognition of the importance of integrating symbolic and sub-symbolic approaches to enhance AI’s reasoning capabilities. The contribution from this systematic literature review is a well-grounded definition of Meta-Cognition within Neuro-Symbolic AI, a review of the key themes of the literature post the Neuro-Symbolic research explosion from 2020-2024 and an identification of the current gaps in the literature of Neuro-Symbolic AI.

We found that the majority of the research efforts in between 2020-24 were concentrated in the areas of learning and inference, with a significant portion also dedicated to logic and reasoning, as well as knowledge representation. These areas have seen substantial advancements, with innovative projects and methodologies pushing the boundaries of what AI systems can achieve in terms of understanding, reasoning, and generating human-like responses. However, our review also identifies several critical gaps in the current literature. Despite the substantial progress in learning and inference, there remains a relative sparseness of research focused on explainability and trustworthiness. This gap is particularly concerning given the increasing deployment of AI systems in real-world applications, where transparency and reliability are paramount. Moreover, the intersection of the four main research areas—learning and inference, logic and reasoning, knowledge representation, and explainability and trustworthiness—reveals a significant opportunity for interdisciplinary work. The density of studies that effectively combine these domains indicates the field is generally well integrated. The most underrepresented area in our review is Meta-Cognition. This emerging field, which involves systems' capacity to monitor, evaluate, and adjust their own reasoning and learning processes, holds great potential for advancing AI towards more autonomous and adaptable intelligence. The few existing studies in this domain suggest promising directions, but much more work is needed to develop robust frameworks and practical implementations of Meta-Cognitive architectures.

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